

Task 1 : Theory and Cipher Design

(a) Confusion, diffusion and avalanche effect

Confusion:

Confusion makes the relationship between the encryption key and the ciphertext complex, so that it is difficult for an attacker to determine the key from the ciphertext. In AES, **SubBytes** contributes to confusion by replacing bytes using a nonlinear substitution table called the S-box.

Diffusion:

Diffusion spreads the effect of one plaintext bit across many ciphertext bits. This makes patterns in the original message difficult to recognize. In AES, **ShiftRows** and **MixColumns** provide diffusion by rearranging and mixing the data across the state.

Avalanche effect:

The avalanche effect means that a small change in the input, such as changing one bit of plaintext, should cause many bits of the ciphertext to change. AES demonstrates this through its repeated rounds of substitution, shifting, mixing, and key addition. Ideally, changing one bit should change approximately half of the ciphertext bits.

(b) DES vs AES

Feature	DES	AES
Block size	64 bits	128 bits
Key size	56 effective bits	128, 192 or 256 bits
Number of rounds	16	10, 12 or 14
Cipher structure	Feistel network	Substitution-permutation network
Present-day security	Insecure because its key is too small	Secure when properly implemented with an appropriate key size

(c) ECB, CBC and CTR

ECB (Electronic Codebook):

Each plaintext block is encrypted independently. The same plaintext block produces the same ciphertext block when the same key is used. This can reveal patterns, making ECB unsuitable for most applications.

CBC (Cipher Block Chaining):

Each plaintext block is combined with the previous ciphertext block before encryption. CBC requires an **IV (Initialization Vector)** so that identical plaintext blocks do not produce identical ciphertext blocks at the beginning.

CTR (Counter):

CTR encrypts a counter value to produce a keystream, which is XORed with the plaintext. It does not require padding and can encrypt data of arbitrary length.

The purpose of IV/Nonce:

An IV or nonce provides a fresh starting value so that encrypting the same plaintext with the same key does not normally produce the same ciphertext.

Important misuse to avoid:

A nonce must **never be reused with the same key in CTR mode**. Reusing it can expose relationships between the plaintexts and seriously weaken security